Detailed Description of the Modified IEEE 14-Bus Test System

Technical information supporting Learning Dynamic Security Regions of Power Systems with Koopman Mode Decomposition

1. Purpose and Scope

This document provides the test-system and dataset-construction details supporting the numerical study in the main paper. It supplements the concise description therein and is intended to facilitate reproduction of the numerical study. It records the system base, dynamic component models and parameters, base-case operating data, contingency sequence, operating-point sampling, trajectory generation, and training/test split used for KMD-based damping labeling and machine-learning-based dynamic security region (DSR) construction.

2. System Configuration

The study uses a modified IEEE 14-bus power-system model implemented in the Power System Analysis Toolbox (PSAT). The system configuration is summarized in Table S1. The system base is 100 MVA and 60 Hz. Five fourth-order synchronous generators are connected to Buses 1, 2, 3, 6, and 8. Generators 1–5 in the parameter tables correspond to these buses in the same order. The network contains 21 branches, including two parallel circuits between Buses 1 and 2.

Table S1. System-level configuration

Item	Specification
Simulation platform	PSAT
Network	Modified IEEE 14-bus system
System base	100 MVA, 60 Hz
Synchronous generators	Five fourth-order machines at Buses 1, 2, 3, 6, and 8
Operating-point variables	Active-power outputs at Buses 2 and 6
Fixed quantities	Other generator and load setpoints and dynamic-model parameters; slack active power adjusts to the power-flow solution

The network parameters on the 100-MVA system base are listed in Table S2. Branch numbers follow the PSAT case-file order. The columns r, x, and b give the series resistance, series reactance, and shunt susceptance, respectively. The columns k and tap retain the corresponding PSAT model parameters.

Table S2. Network branch parameters

Row	From	To	*r*	*x*	*b*	*k*	*tap*
1	2	5	0.05695	0.17388	0.034	0.0	0.0

Row	From	To	*r*	*x*	*b*	*k*	*tap*
2	6	12	0.12291	0.25581	0.0	0.0	0.0
3	12	13	0.22092	0.19988	0.0	0.0	0.0
4	6	13	0.06615	0.13027	0.0	0.0	0.0
5	6	11	0.09498	0.1989	0.0	0.0	0.0
6	11	10	0.08205	0.19207	0.0	0.0	0.0
7	9	10	0.03181	0.0845	0.0	0.0	0.0
8	9	14	0.12711	0.27038	0.0	0.0	0.0
9	14	13	0.17093	0.34802	0.0	0.0	0.0
10	7	9	0.0	0.11001	0.0	0.0	0.0
11	1	2	0.01938	0.05917	0.0528	0.0	0.0
12	3	2	0.04699	0.19797	0.0438	0.0	0.0
13	1	2	0.01938	0.05917	0.0528	0.0	0.0
14	3	4	0.06701	0.17103	0.0346	0.0	0.0
15	1	5	0.05403	0.22304	0.0492	0.0	0.0
16	5	4	0.01335	0.04211	0.0128	0.0	0.0
17	2	4	0.05811	0.17632	0.0374	0.0	0.0
18	5	6	0.0	0.25202	0.0	5.0	0.932
19	4	9	0.0	0.55618	0.0	5.0	0.969
20	4	7	0.0	0.20912	0.0	5.0	0.978
21	8	7	0.0	0.17615	0.0	1.304348	0.0

3. Dynamic Models and Parameters

The synchronous-machine parameters are listed in Tables S3 and S4 for the *d* and *q* axes, respectively. Per-unit machine parameters use the corresponding device ratings: 615, 60, 60, 25, and 25 MVA at Buses 1, 2, 3, 6, and 8, respectively. The inertia entry is $M=2H$. These tables include the machine parameter entries in the PSAT model; only the parameters required by the selected fourth-order model enter its dynamics. AVR II controllers are installed on all five generators, with their parameters listed in Table S5. TG I controllers are installed at Buses 1 and 2 (Generators 1 and 2), with their parameters listed in Table S6.

Table S3. Detailed dynamic parameters of synchronous generators - *d* axis

Gen. No.	x_d	x'_d	x''_d	T'_{d0}	T''_{d0}	$M=2H$
1	0.8979	0.2998	0.23	7.4	0.03	10.296
2	1.05	0.185	0.13	6.1	0.04	13.08
3	1.05	0.185	0.13	6.1	0.04	13.08
4	1.25	0.232	0.12	4.75	0.06	10.12

Gen. No.	x_d	x'_d	x''_d	T'_{d0}	T''_{d0}	$M=2H$
5	1.25	0.232	0.12	4.75	0.06	10.12

Table S4. Detailed dynamic parameters of synchronous generators - *q* axis

Gen. No.	x_q	x'_q	x''_q	T'_{q0}	T''_{q0}	D
1	0.646	0.646	0.4	0	0.033	0
2	0.98	0.36	0.13	0.3	0.099	0
3	0.98	0.36	0.13	0.3	0.099	0
4	1.22	0.715	0.12	1.5	0.21	0
5	1.22	0.715	0.12	1.5	0.21	0

Table S5. Automatic voltage regulator parameters (AVR II)

Gen. No.	K_a	T_a	K_f	T_f	T_e	T_r	A_e	B_e
1	200	0.02	0.002	1	0.14	0.001	0.0006	0.9
2	20	0.02	0.001	1	0.14	0.001	0.0006	0.9
3	20	0.02	0.001	1	0.14	0.001	0.0006	0.9
4	20	0.02	0.001	1	0.1	0.001	0.0006	0.9
5	20	0.02	0.001	1	0.1	0.001	0.0006	0.9

Table S6. Governor parameters (TG I)

Gen. No.	R	T_s	T_e	T_1	T_2	T_3
1	0.02	0.1	0.45	0	12	50
2	0.02	0.1	0.45	0	12	50

4. Base-Case Operating Data

Tables S7 and S8 list the base-case load demands and generator operating data. The active-power outputs at Buses 2 and 6 are varied over the ranges in Section 6. The remaining generator active-power setpoints, load settings, and generator voltage setpoints are held fixed. Bus 1 is the slack bus; its active-power output adjusts to balance generation, demand, and network losses. The load component at Bus 2 is disabled in the PSAT model, although its specified demand values are included in Table S7. Generator 1 is the slack machine, with a voltage setpoint of 0.9796 p.u. Its active-power output is determined by the power-flow solution for each sampled

operating point. The active-power entries for Generators 2 and 4 in Table S8 are base-case values and are replaced by sampled values during dataset construction.

Table S7. Detailed load parameters

Bus No.	P_L (p.u.)	Q_L (p.u.)
11	0.0578	0.0333
13	0.1945	0.1007
3	1.5445	0.3177
5	0.1340	0.0224
2	0.4122	0.1749
6	0.1445	0.1355
4	0.6440	0.0633
14	0.2330	0.0749
12	0.0868	0.0266
10	0.1404	0.0972
9	0.5198	0.2415

Table S8. Detailed generator operating data

Gen. No.	P_G (p.u.)	V_G (p.u.)
2	0.9447	0.9849
3	0.2567	1.0520
4	0.0753	0.9883
5	0.0021	0.9768

5. Contingency Definition

A single fixed disturbance sequence, summarized in Table S9, is applied to every sampled operating point. The learned DSR is therefore contingency-specific and represents the operating conditions whose post-disturbance responses satisfy the selected damping requirement. At t=1.2 s, branch 12, connecting Buses 3 and 2, is opened at its Bus-3 terminal and remains out of service for the rest of the 50-s simulation. Both parallel circuits between Buses 1 and 2 remain in service.

Table S9. Disturbance sequence

Time	Event
t = 0 s	A three-phase fault is applied at Bus 4.
t = 0.02 s	The three-phase fault is cleared.
t = 1.2 s	Branch 12 (Bus 3–Bus 2) opens at the Bus-3 terminal.

6. Operating-Point Sampling and DSR Parameter Space

The active-power outputs of the generators at Buses 2 and 6 are selected as the operating-point variables. Each operating point is represented by the pair (P_{G2}, P_{G6}). The sampling settings are summarized in Table S10. A total of 10,000 operating points are sampled independently and uniformly, with P_{G2} in [0, 1.4] p.u. and P_{G6} in [0, 1.0] p.u. This two-dimensional parameterization permits direct visualization of the damping-performance boundary and the learned DSR.

All dynamic-model parameters and network settings are identical across the sampled operating points before the specified contingency. The remaining generator setpoints and load settings retain their base-case values, while the slack-generator output is recalculated for each point.

Table S10. Operating-point sampling

Quantity	Value
Operating-point vector	(P_{G2}, P_{G6})
P_{G2} range	[0, 1.4] p.u.
P_{G6} range	[0, 1.0] p.u.
Number of samples	10,000 sampled points; 9,999 valid samples
Sampling rule	Independent uniform sampling

7. Time-Domain Simulation and KMD Observables

For each sampled operating point, a power-flow solution initializes a 50-s time-domain simulation in PSAT. The trajectory-generation settings are summarized in Table S11. The nominal integration step is 0.05 s. PSAT may reduce the step around fault events or when required for convergence; the trajectory output includes the corresponding time stamps. Shorter observation windows, including 15 s, are extracted from the simulated responses for the observation-window comparisons.

The rotor-angle signals are measured relative to the slack-machine angle, and the rotor-speed deviations are measured relative to each machine's initial speed. Before KMD computation, the angle signals are also expressed as changes from their initial values. The angle and speed-deviation groups are normalized separately: each group is divided by its maximum absolute value across generator channels and time samples within the observation window. These normalized signals form the multivariate KMD observables.

Table S11. Trajectory-generation settings

Item	Specification
Simulation duration	50 s per operating point
Nominal integration step	0.05 s; PSAT may reduce it internally
Recorded signals	Relative rotor angles and rotor-speed deviations
Preprocessing	Separate normalization of angle and speed-deviation groups before KMD
Sample structure	Operating-point vector plus post-disturbance trajectory

8. Dataset Partition and Intended Use

KMD provides damping labels for the sampled operating points. KNN-based correction is then applied before ML-based DSR learning. The labeled samples are divided into 80% for training and 20% for testing, and the same partition is retained across the reported comparisons. The training subset is used to learn the DSR, while the held-out test subset is used to evaluate operating-point classification performance.

9. Dataset Contents

Each sample consists of an operating-point vector, the corresponding generator and load operating data, a multivariate dynamic trajectory, and its time stamps. Of the 10,000 sampled operating points, 9,999 valid samples are used in the reported comparisons. Damping labels and corrected labels are associated with the same operating points so that label generation, label correction, and DSR learning can be evaluated consistently.